

Global Energy Storage

China Next Winners: Powering China's AI Revolution



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This report is part of a series on China Next Winners in which we identify the next generation of industries and corporate winners in China.

China's AI benefits from a structural cost advantage. Leading Chinese models are often priced at a fraction of comparable US models while maintaining competitive intelligence scores, resulting in attractive intelligence-per-dollar proposition. This advantage is driven by a combination of more efficient model architectures, lower infrastructure costs and cheaper power, allowing Chinese providers to deliver AI inference at materially lower cost.

We expect hyperscaler electricity consumption to become one of the fastest growing sources of power demand over the next decade. Driven by rising AI adoption, continued expansion of data center infrastructure and China's ambition to narrow the compute gap with the US, we estimate data center electricity demand could increase from 202TWh in 2025 to 1,476TWh in 2035, representing a 24% CAGR. While China currently has only ~15% of US AI compute capacity, the country is rapidly scaling both domestic AI chips and computing infrastructure. Achieving meaningful progress toward US-level compute capacity will require substantial growth in data-center capacity, which in turn creates significant incremental demand for electricity.

Cheap, abundant and scalable power supports China's AI ambitions. China already generates more than twice as much electricity as the US and added over 500GW of new power capacity in 2025 alone. At the same time, China enjoys some of the lowest industrial electricity prices globally, while major AI hubs in Inner Mongolia, Ningxia and Gansu benefit from some of the world's lowest data center power tariffs. China also combines low cost renewables with competitively built nuclear generation, providing both affordability and reliability for AI workloads.

Data center power demand will be increasingly supplied by renewable energy. Through "East Data, West Computing", green-data-center initiatives and the emerging "Computing-Power Coordination" framework, policymakers are increasingly aligning compute growth with power infrastructure. New national hub data centers are required to source more than 80% of their electricity from green power, while leading operators have committed to ambitious renewable-energy targets of 100% renewable by 2030.

We see the most compelling opportunities in grid infrastructure and energy storage. Massive planned investment in UHV transmission and grid infrastructure should benefit equipment suppliers, while rising renewable penetration and increasing AI power demand reinforce the need for battery storage. Among our coverage, **CATL** is a key beneficiary through its leadership in grid-scale storage systems, while **Sungrow** is positioned across renewables, power conversion, storage and grid integration.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	14 Sep 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	337.11	800.00	(27.4)%	CNY	16.14	21.95	28.77	20.9	15.4	11.7
3750.HK (CATL)	O	HKD	549.00	770.00	(4.4)%	CNY	16.14	21.95	28.77	29.1	21.4	16.3
247540.KS (EcoPro BM)	U	KRW	103,900	140,000	(35.2)%	KRW	403.00	854.00	1,963.00	257.8	121.7	52.9
051910.KS (LG Chem)	M	KRW	268,000	312,000	(32.1)%	KRW	13,258.70	2,693.22	28,743	(20.2)	99.5	9.3
373220.KS (LGES)	M	KRW	351,500	351,000	(24.6)%	KRW	(4,580.94)	(2,364.33)	10,225	(76.7)	(148.7)	34.4
003670.KS (Posco Future M)	U	KRW	178,900	116,000	11.8%	KRW	376.72	379.39	781.72	474.9	471.6	228.9
006400.KS (SDI)	M	KRW	532,000	520,000	140.6%	KRW	(9,933.80)	1,751.66	18,577	(53.6)	303.7	28.6
300274.CH (Sungrow)	O	RMB	84.98	151.00	(61.3)%	RMB	6.55	6.93	7.95	13.0	12.3	10.7
002466.CH (Tianqi Lithium)	O	CNY	42.97	80.00	(28.6)%	CNY	0.28	3.90	7.75	152.3	11.0	5.5
9696.HK (Tianqi Lithium)	O	HKD	31.78	65.00	(45.5)%	CNY	0.28	3.90	7.75	96.3	7.0	3.5
034020.KS (Doosan Enerbility)	O	KRW	87,000	100,000	24.5%	KRW	132.34	691.11	1,249.76	657.4	125.9	69.6
ASIA			1,938.16									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

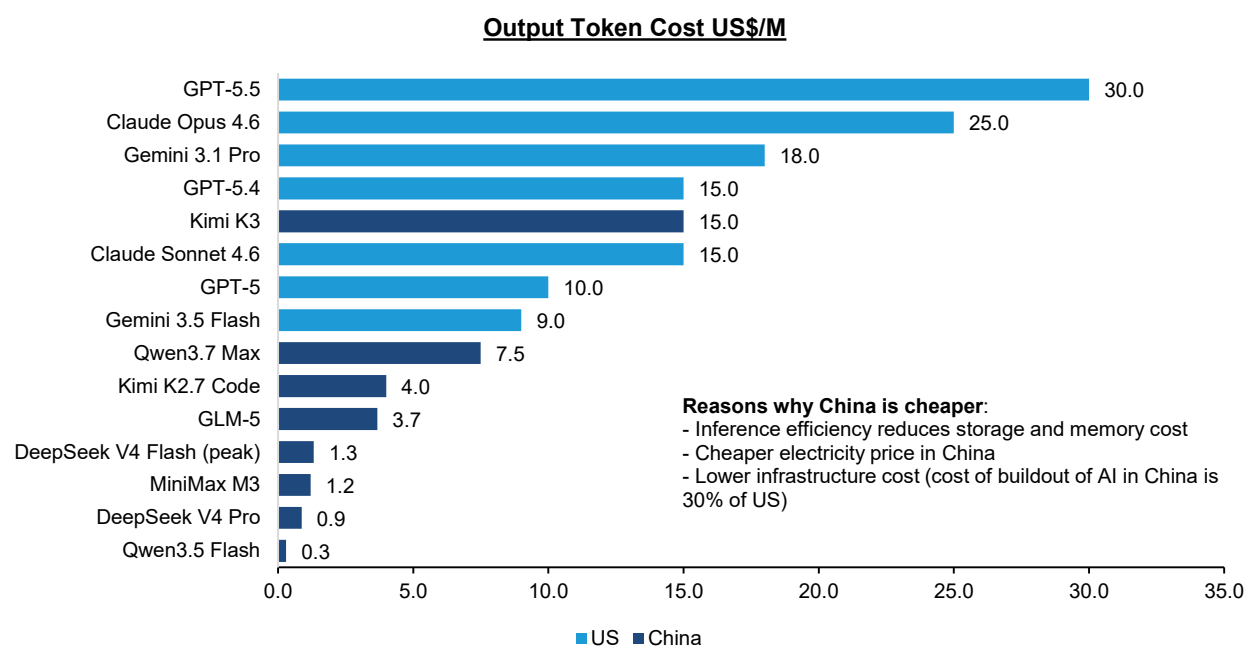
We believe China's AI ecosystem is emerging as a key beneficiary of the country's unique advantages in power, infrastructure and policy support. Leading Chinese AI models are increasingly delivering competitive intelligence at materially lower cost, supported by efficient architectures, lower infrastructure costs and some of the world's lowest power tariffs. As China seeks to narrow the compute gap with the US, we estimate data-center electricity demand could rise from 202TWh in 2025 to 1,476TWh by 2035, making hyperscalers one of the fastest growing sources of power demand. Unlike many markets, China combines abundant and low-cost electricity, rapid capacity expansion, competitive power supply chains, and policy initiatives that increasingly align compute growth with power infrastructure. In our view, this creates a multi-year investment opportunity across the AI power value chain, particularly in grid infrastructure and energy storage, with CATL and Sungrow among the most direct beneficiaries of rising investment in transmission, renewable integration and power reliability required to support large scale AI deployment.

DETAILS

FROM CHEAP POWER TO CHEAP TOKENS

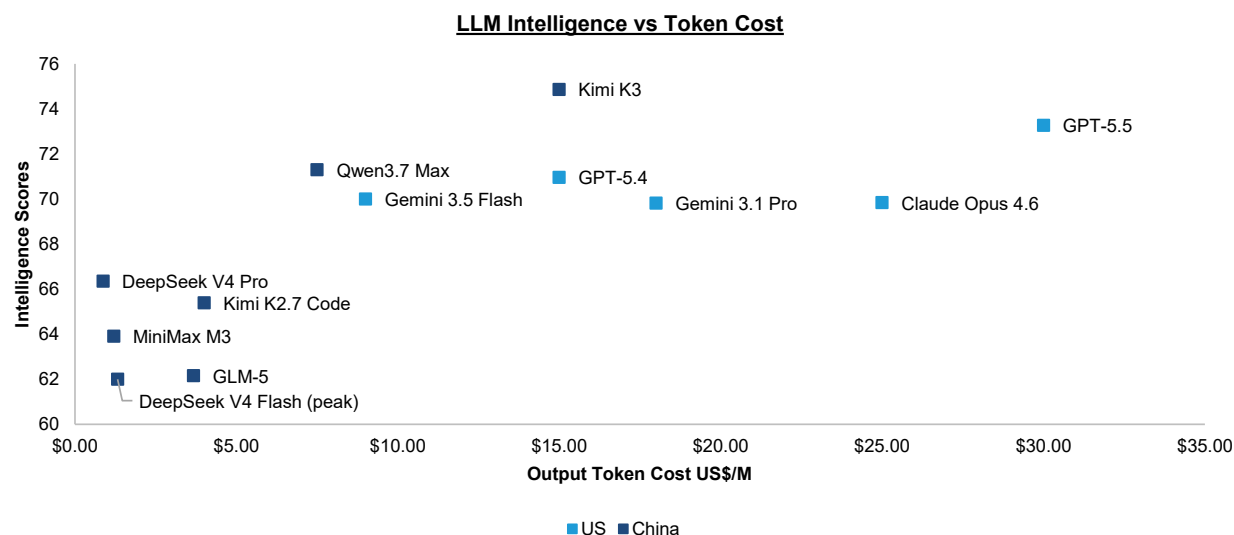
China's AI industry is increasingly competing on economics rather than model capability alone. While leading US models continue to dominate the frontier of performance, a growing number of Chinese models are delivering comparable levels of intelligence at a fraction of the cost. The gap is particularly striking in inference pricing, where leading Chinese models are priced significantly below leading US models. More importantly, the intelligence-versus-cost comparison suggests that Chinese models are not simply competing on price. Many are delivering similar performance at substantially lower token costs, creating a compelling value proposition for enterprises and developers as AI adoption increasingly shifts from model training toward large-scale inference.

EXHIBIT 1: **China LLM's have a token cost which is significantly lower than US**



Source: Company data, Bernstein analysis

Importantly, lower pricing does not appear to come at the expense of model quality. When comparing model intelligence scores against output token costs, several leading Chinese models are capable of delivering intelligence levels that are broadly comparable to frontier US models while charging substantially lower prices per token. This suggests that Chinese AI developers are increasingly optimizing for intelligence per dollar. As AI adoption shifts from training a handful of frontier models to serving large-scale inference workloads, cost-performance efficiency is likely to become an increasingly important competitive metric. In this environment, China's advantages in power costs, infrastructure economics and model efficiency could translate into a structurally lower cost of delivering AI services at scale.

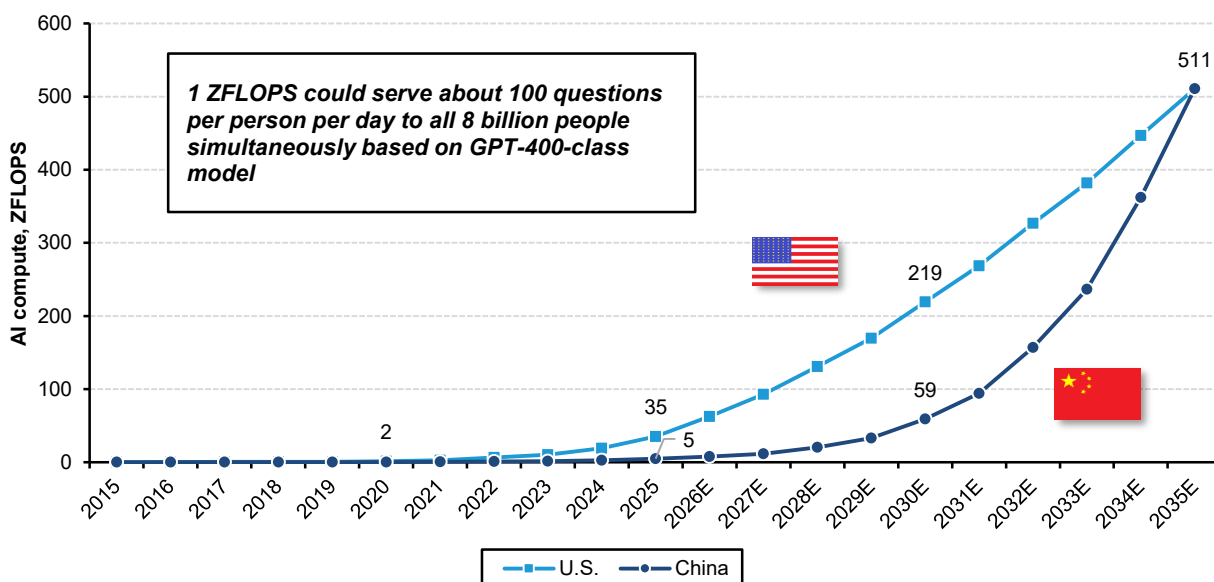
EXHIBIT 2: Importantly, lower pricing does not appear to come at the expense of model quality

BenchLM intelligence score is a relative, weighted measure of a model's broad capability. It combines performance across agentic tasks (22%), coding (20%), reasoning (17%), knowledge (12%), multimodal and grounded tasks (12%), multilingual capability (7%), instruction following (5%) and mathematics (5%).
Source: Company data, BenchLM.ai, Bernstein analysis

HOW MUCH ELECTRICITY WILL CHINA'S AI BUILDOUT REQUIRE?

If China is to reach the same compute power as the US, the first question is, what is the expected growth in US compute power? Based on an outlook that assumes chip efficiency improvements follow Nvidia's technological trajectory in recent years, and incorporating data center power demand expectations widely tracked by the market (BNEF), we project U.S. AI compute capacity will increase by 15x by 2035 over the next 10 years. Over the past five years, AI compute in the U.S. has expanded from roughly 2 ZFLOPs to 35 ZFLOPs, an 86% CAGR. Based on current capex plans and an extrapolation of growth trends we forecast the US compute power will reach 219 ZFLOPS by 2030 and 511 ZFLOPS by 2035. Today, China's AI compute stands at around 5 ZFLOPs, roughly 15% of the U.S. By 2030 we expect China will still be significantly behind the U.S., as the chip performance and AIDC investment both need time to ramp up. The question we ask in this report is what would it take in terms of capex, semiconductor and power to reach the same capacity as the US today, which is 511 ZFLOPS by 2035, or 100x more than China's current compute capacity.

EXHIBIT 3: AI compute comparison between the US and China. How long it may take China to catch up?

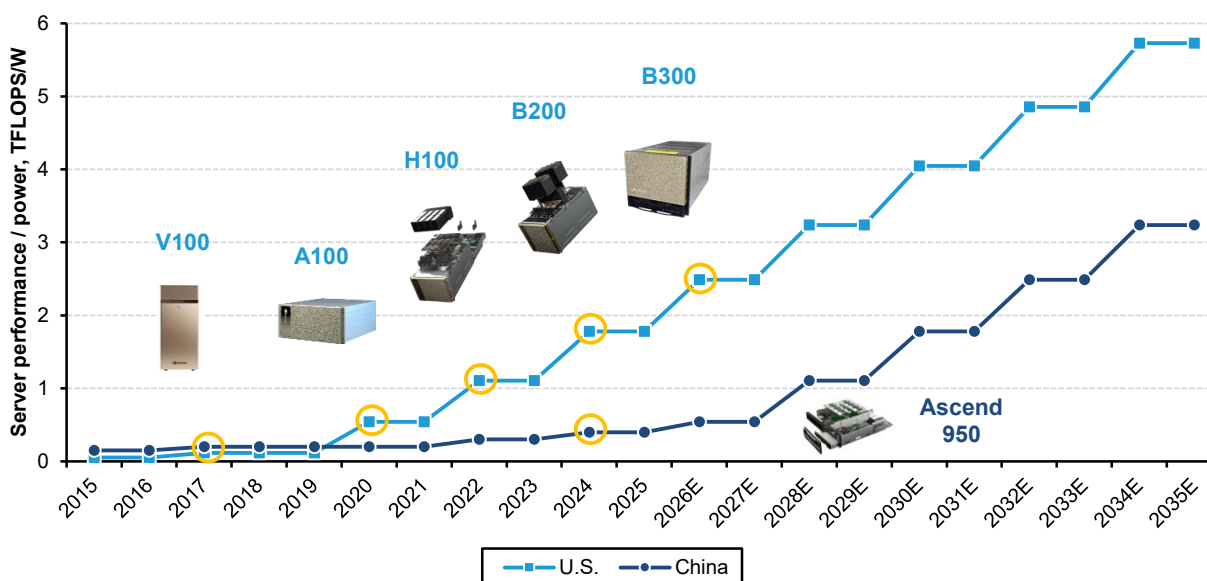


The compute power is measured in FP8 sparse precision

Source: Company data, BNEF, Bernstein analysis and estimates

The amount of power to reach this level of compute is enormous, but fundamentally depends on the progress of China's domestic semiconductor industry. Our assumption is that the US will continue to limit the supply of high-end semiconductors for China and that China will have to rely primarily on domestic AI chips designed and manufactured locally. In 2026, leading Chinese accelerators (represented by the Huawei Ascend 950) offer only 0.5 TFLOPS/W (the power consumption is calculated using total AIDC, not just the rack), which is about one-quarter of the efficiency of U.S. chips (c. 2.5 TFLOPS/W). However, based on our estimation on the gap of chip performance, we believe the gap between China and US would become smaller, with China potentially improving to >50% vs US by 2035. As shown in the past few years, even with tight export control, SMIC has been able to maintain a ~6 year tech gap compared to TSMC. We believe with domestic equipment and process/structure innovation, the gap won't become bigger in the next decade. Based on our analysis, we estimate the China's chips could be equivalent to the best chips available in the US today in terms of performance by 2030. While we expect China to remain behind the U.S. in chip efficiency through 2035, the gap should narrow meaningfully. By 2035, we estimate Chinese chips could reach 3TFLOPS/W which is more than half of the U.S. performance level on a per watt basis.

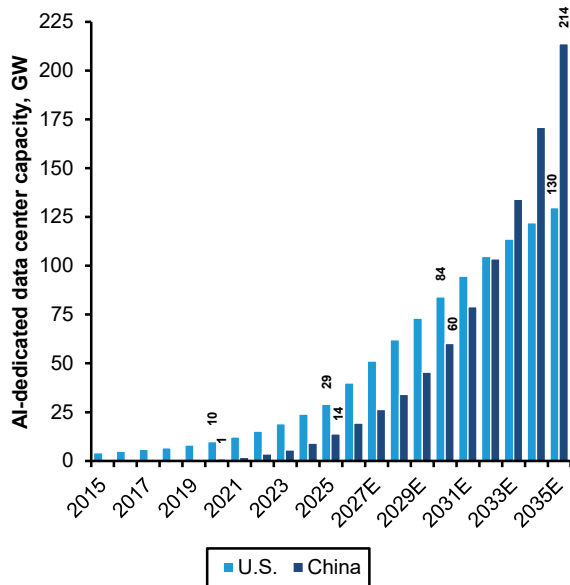
EXHIBIT 4: Today, leading Chinese accelerators (represented by the Huawei Ascend 950) offer only 0.5 TFLOPS/W which is about one-quarter of the efficiency of U.S. chips. But the gap should narrow



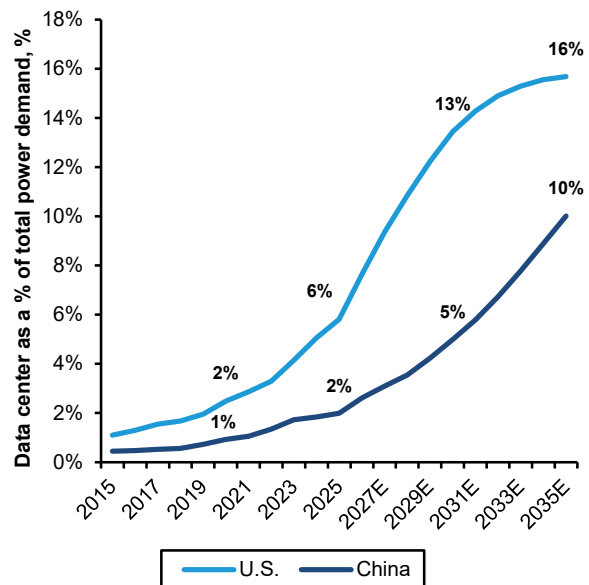
Source: Company data, BNEF, Bernstein analysis and estimates

Given lower power efficiency and smaller computing power per chip, China will have to produce more chips and build larger hyperscalers to match the US in compute power, which also consumes more energy for the same computing power output. Currently, the US has about twice, or 29GW of data center capacity compared to 14GW in China. Given lower performance per W for the domestic chips in Chinese data centers the compute power today is less than 20% of the US. We expect US data center capacity to grow to 84GW by the end of the decade and 214GW by 2035 (Exhibit 5). To reach a level of AI compute comparable to that of the U.S., we estimate that China would need to expand its data center capacity to 214GW by 2035 which is a 15x increase from current levels (Exhibit 5). By 2035, China data center capacity would be 1.6x the US in GW. Considering today's gap, this reflects the progress in China semiconductor performance.

The power demand will be significantly higher in China, but given China's much larger production of power, the percent of total power demand which will come from hyperscalers will be less. By 2035 data centers are planned to account for about 10% of China's total electricity demand, compared with 16% in the U.S. China's meaningful power demand acceleration from AI data centers is likely to come later in the next decade, during which we expect their share of total power consumption to rise from 5% to 10% (Exhibit 6). In contrast, the U.S. is set for a more front loaded surge: over the next five years, U.S. data centers' share of national electricity demand is projected to increase from 6% to 13%. A question is whether this will be achievable in the US and Elon Musk pointed out in Davos this year, it could be that the 'US is producing more chips than they can turn on'.

EXHIBIT 5: To reach parity with U.S. AI compute, China would need to expand data center capacity to 214GW

Source: Company data, BNEF, Bernstein analysis and estimates

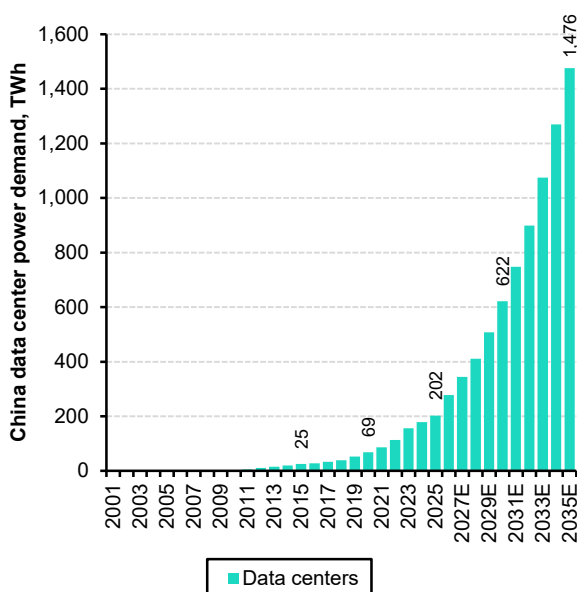
EXHIBIT 6: The data center will account for 10% of total power demand in China by 2035, vs. 16% for the U.S.

Source: Company data, BNEF, Bernstein analysis and estimates

We assume that China will match the U.S.'s total compute power by 2035. Today, China's AI compute is about 5ZFLOPS, roughly 15% of the U.S. level. In theory, catching up is possible, given the huge efforts ongoing, but there are also real constraints. For the U.S., power supply is a major bottleneck. For China, the bottleneck is advanced, high-efficiency AI chips, both in design and manufacturing. This raises an important question. If China relies heavily on lower-efficiency domestic chips, will power availability become a limiting factor?

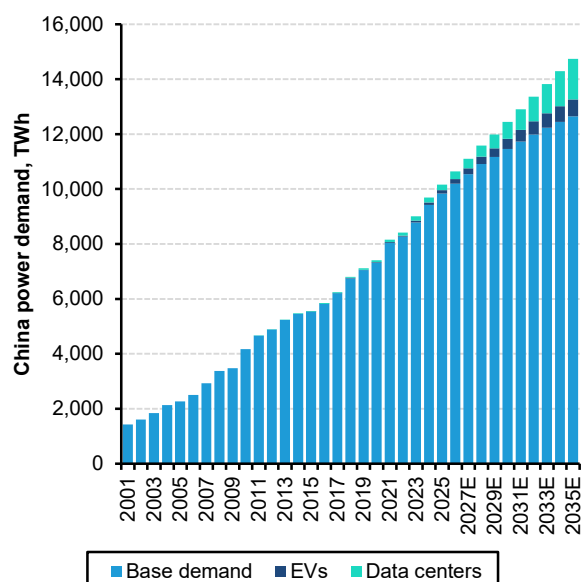
To explore this, we develop a scenario where China reaches U.S. equivalent AI compute capacity by 2035. In other words, China's AI compute rises from 5 ZFLOPS today to 59 ZFLOPS in 2030 (a 10x increase) and 511 ZFLOPS in 2035 (a 100x increase). While this seems dramatic, we assume most of this scaling comes from higher chip efficiency. For new chips, China's power efficiency improves from 0.4 TFLOPS/W today to 3.2 TFLOPS/W in 2035 (8x improvement). By comparison, U.S. chip efficiency is assumed to rise from 1.8 TFLOPS/W to 5.7 TFLOPS/W over the same period.

EXHIBIT 7: To match the US AI compute, we expect China data center power demand to grow from 202TWh in 2025 to 1476TWh in 2035 (+24% CAGR) ...



Source: BNEF, Bernstein analysis and estimates

EXHIBIT 8: ... And total power demand to increase from today's 10163TWh to 14741TWh in 2035, representing a CAGR of +3.8%



Source: BNEF, SNE Research, Bernstein analysis and estimates

Under these assumptions, China's data-center electricity demand grows from 202TWh in 2025 to 1,476TWh in 2035, a 24% CAGR. Total national electricity demand grows from 10,163TWh today to 14,741TWh in 2035, or 3.8% CAGR. Data centers become a meaningful driver of overall growth, adding 1,274 TWh of demand and accounting for 28% of China's total electricity demand growth over the next decade. However, the contribution from data center demand is still smaller in relative terms than in the U.S., where AI-driven load growth (624TWh) could represent more than half of future demand growth.

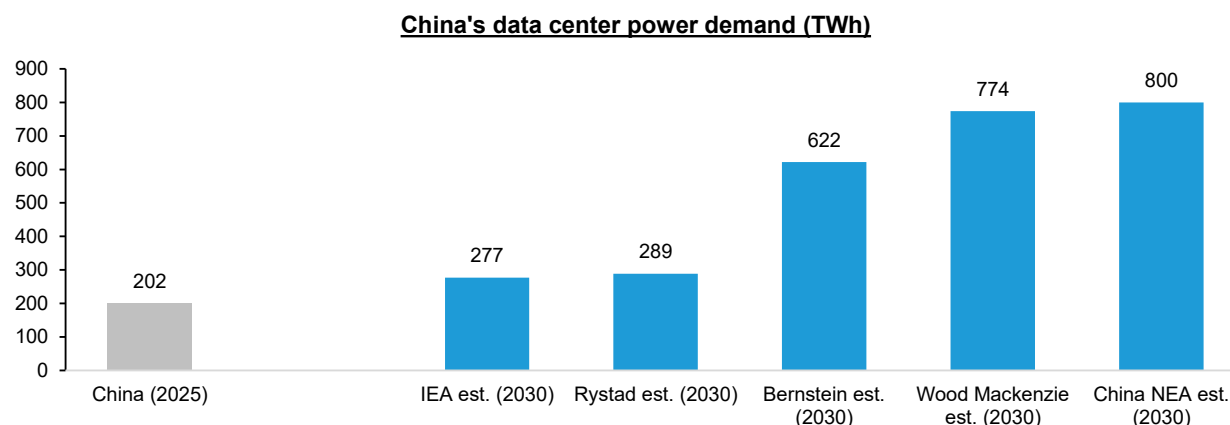
EXHIBIT 9: Assumptions and projections underlying our estimates of China data center power consumption and AI compute demand

Data center power and AI compute (China)	2020	2021	2022	2023	2024	2025	2030E	2035E	CAGR		
									20-25	25-30	30-35
Data center power demand, TWh	69	86	113	156	178	202	622	1,476	24%	25%	19%
- Data center IT equipment power draw, TWh	45	57	76	106	123	141	468	1,200	26%	27%	21%
Data center installed capacity, GW	11	14	19	25	28	32	93	238	23%	24%	21%
- AI dedicated capacity, GW	1	2	4	6	9	14	60	214	69%	34%	29%
AI compute, ZFLOPS	0	0	1	2	3	5	59	511	89%	66%	54%
Fleet efficiency, TFLOPS/W, cumulative avg.	0.2	0.2	0.2	0.3	0.3	0.3	1.0	2.4	12%	23%	19%
Fleet efficiency, TFLOPS/W, new added	0.2	0.2	0.3	0.3	0.4	0.4	1.8	3.2	15%	35%	13%

Source: Company data, BNEF, Bernstein analysis and estimates

China's National Energy Administration (NEA) has set an official trajectory with computing center electricity consumption expected to rise to 800TWh by 2030 which could account for 6% of national electricity consumption. Several other industry groups have pegged 2030 data center power demand at around 300-800TWh compared to our 622TWh. The spread reflects real uncertainty about how fast announced compute capacity converts into grid-connected, operating capacity.

EXHIBIT 10: Industry groups have pegged 2030 data center power demand at around 300-800TWh compared to our 622TWh



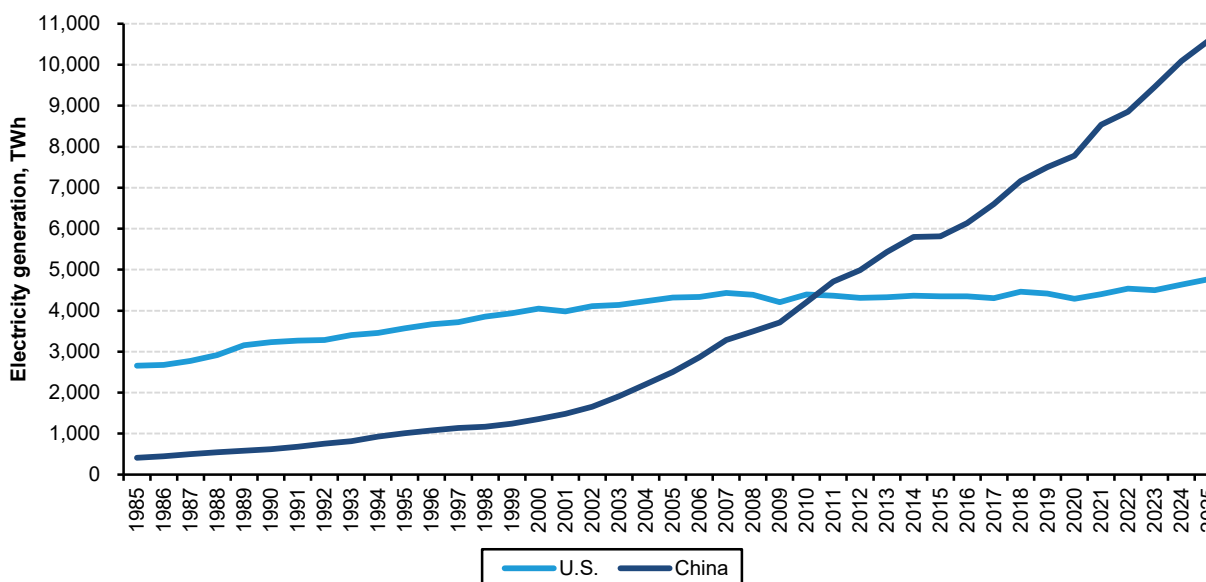
Source: BNEF, NEA, Rystad, IEA, Wood Mackenzie, Bernstein estimates and analysis

CHINA'S POWER ADVANTAGE

1) SCALING POWER CAPACITY

China is the world's largest producer and consumer for electricity, consuming over 10,000TWh annually. This is more than double the consumption of the United States which produces about 5,000TWh annually. Last year, China became the first country in the world to consume over 1,000TWh consumption in a single month, which is comparable to the annual power consumption of Japan. Today electricity accounts for about 30% of China's total energy mix, significantly higher than the global average of 21%. Despite this already high reliance on electricity, China is rapidly evolving into being an 'electrostate' as it seeks to focus on domestic power supply growth.

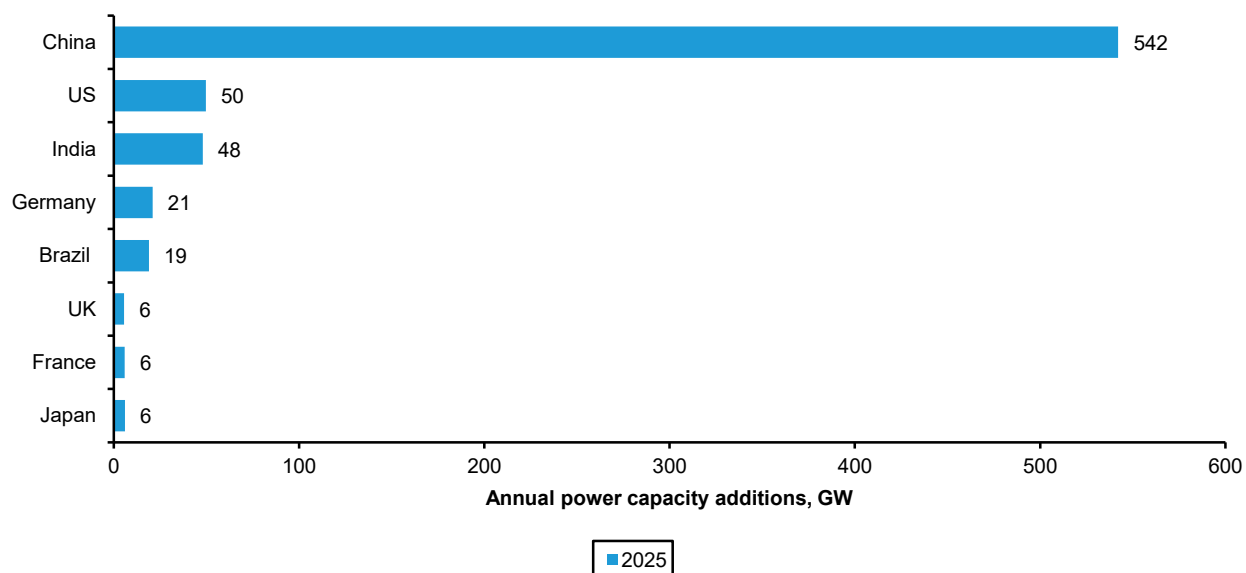
EXHIBIT 11: China's total power generation surpassed that of the U.S. in 2010, and today it produces roughly twice as much. China's ambition is to become the world's leading Electro-state with 50% of energy from electricity by 2050



Source: BP, Bernstein analysis

No other country adds as much power capacity as China every year. According to our estimates, China successfully added a staggering 500GW of power capacity in 2025 (dominated by solar and wind) which is almost an order of magnitude more than the US, India or the EU. China's total solar manufacturing capacity of 1000GW and wind capacity is over 200GW, which implies that China with 1200GW of manufacturing capacity still has scope to raise power capacity additions even further.

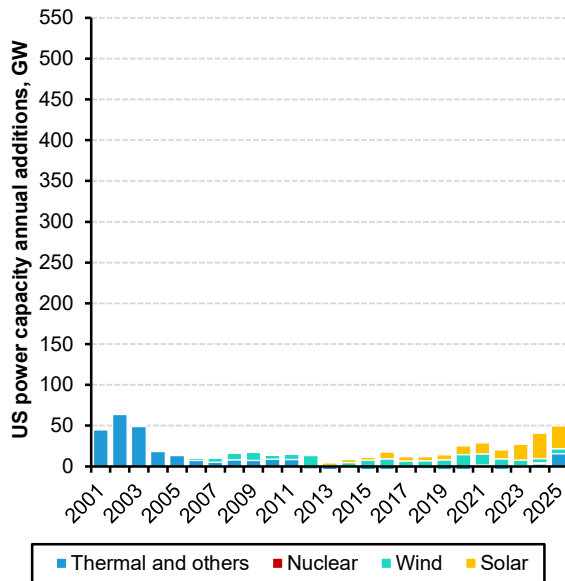
EXHIBIT 12: China is adding more power capacity than the rest of the world combined and 10x more than the US



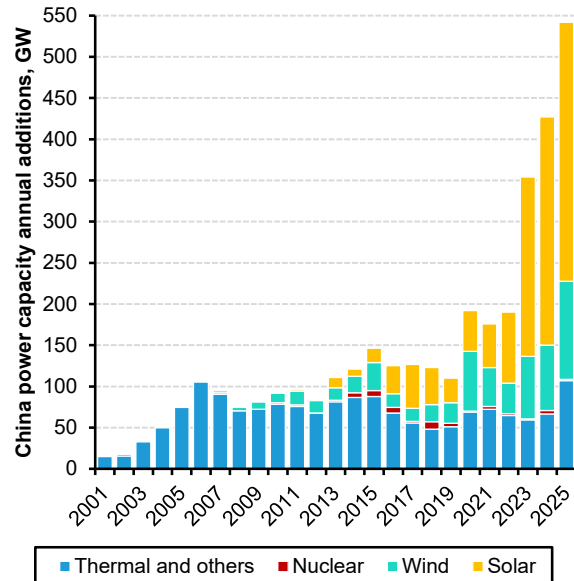
Source: Government data, Bernstein analysis

Comparing the US and China highlights the vast differences in growth of power supply. Over the past 25 years, there has only been 1 year in which the US added more than 50GW of power capacity. The average increase in capacity per year has been closer to 20GW. In 2025, the U.S. added 50GW of new capacity, including 34GW from solar and wind. Although the Trump administration has been negative on solar and wind, cutting key elements of policy support, renewables are increasingly becoming the fastest and most scalable solution to meet stronger-than-expected power demand growth, particularly from data centers. While power demand in the US has been increasing over recent years, looking back over the past decade, power demand has been relatively flat at c. 4000TWh annually, with nuclear, solar and wind playing an increasingly important role.

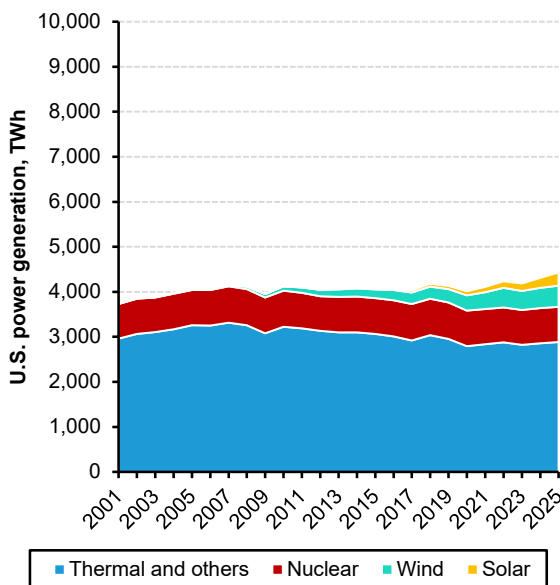
At the same time, China's power capacity additions have surged over the past 25 years. In the past year, China has added over 500GW of power, 314GW of solar capacity and 119GW of wind capacity, which is more than the rest of the world combined and ten times more than the US. Renewables are playing an increasingly critical role in meeting incremental power demand. In 2025, roughly 80% of all new power capacity came from renewable sources (though their share of actual generation is lower due to capacity-factor differences). China power demand has increased in line with power capacity additions. Total power demand has increased by an order of magnitude over the past decade from 1000TWh to 10,000TWh. As source of low carbon power generation increase, so do the share of renewables which supplied 22% of China's total electricity generation in 2025, producing around 2,300TWh.

EXHIBIT 13: Both thermal power and renewable capacity additions in the US have expanded rapidly since 2024

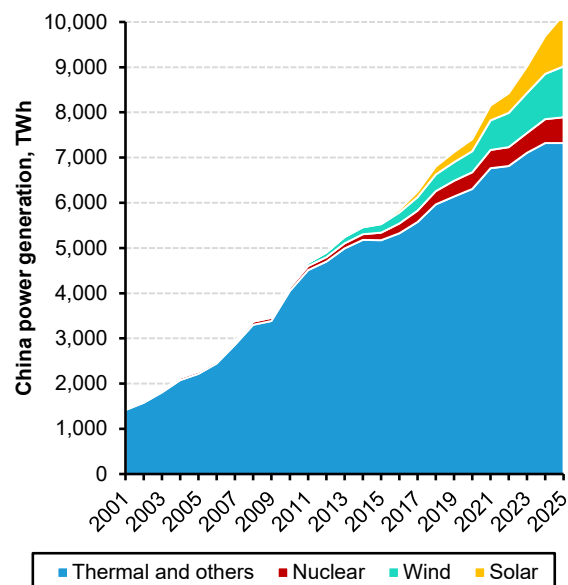
Source: EIA, Bernstein analysis

EXHIBIT 14: The rate of capacity additions of solar, wind, nuclear and hydro has trebled over the past 3 years

Source: China NBS, Wind, Bernstein analysis

EXHIBIT 15: While in the U.S., the total generation from renewables reached 760TWh (17% of total)

Source: EIA, Bernstein analysis

EXHIBIT 16: Renewables accounted for 22% of China's power generation in 2025, totaling 2,300TWh

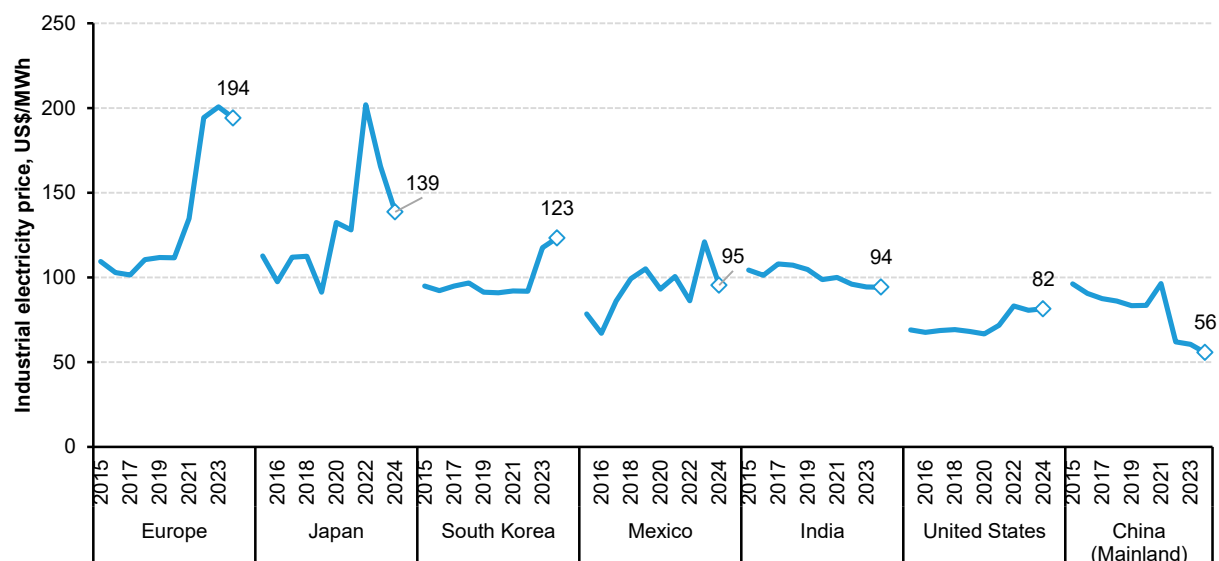
Source: BNEF, Bernstein analysis

2) COST ADVANTAGE

Beyond China's ability to scale rapidly and its highly mature renewables supply chain, another key factor supporting its trajectory toward becoming the world's largest electrostate is cost competitiveness. China enjoys one of the lowest industrial electricity prices among major economies, with average industrial tariffs of approximately US\$56/MWh. This cost advantage is particularly important for AI and hyperscale data centers, where electricity is a major operating expense. Lower power costs reduce the cost of AI training and inference, supporting more competitive token pricing and lowering the barriers to large-scale compute

deployment. China's position as the lowest-cost major industrial power market therefore provides a structural advantage as AI workloads become increasingly power intensive.

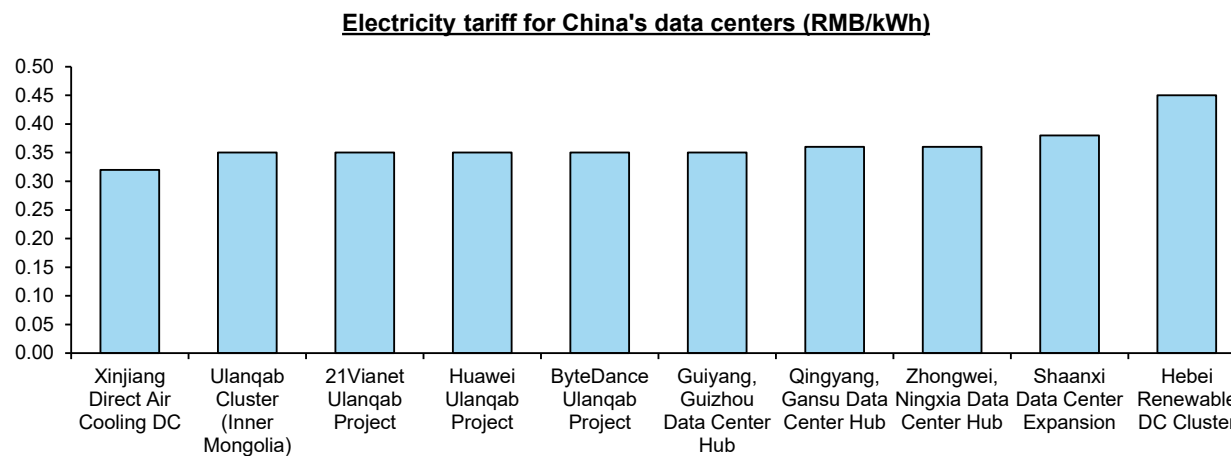
EXHIBIT 17: China enjoys one of the lowest industrial electricity prices among major economies, with average industrial tariffs of approximately US\$56/MWh



Source: IEA, Bernstein analysis

The advantage becomes even more evident when examining power tariffs at China's leading data center hubs. Most major AI and cloud clusters, including Ulanqab, Guizhou, Qingyang and Zhongwei, benefit from electricity tariffs in a relatively narrow range of RMB0.33-0.37/kWh, while some projects in renewable-rich regions such as Xinjiang can access power at close to RMB0.32/kWh. These locations sit at the center of China's "East Data, West Computing" strategy, which colocates compute-intensive workloads with abundant low-cost renewable resources. As a result, China is able to offer both low electricity prices and large-scale power availability, creating a favorable environment for AI data-center expansion while supporting lower-cost AI services relative to many competing markets.

EXHIBIT 18: Most major AI and cloud clusters, including Ulanqab, Guizhou, Qingyang and Zhongwei, benefit from electricity tariffs in a relatively narrow range of RMB0.33-0.37/kWh

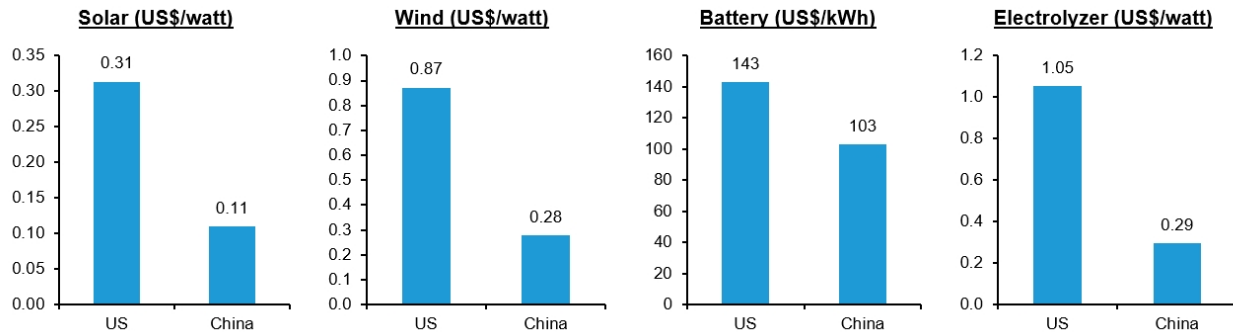


Source: Company data, Woodmac, Bernstein analysis

Costs of renewables in China are significantly lower, often only one-third of those in the U.S., which is critical for achieving long-term net-zero goals. China is the lowest-cost producer across most segments of the renewable value chain and is likely to

maintain this position, supported by structural advantages in energy prices, labor costs, and large-scale manufacturing.

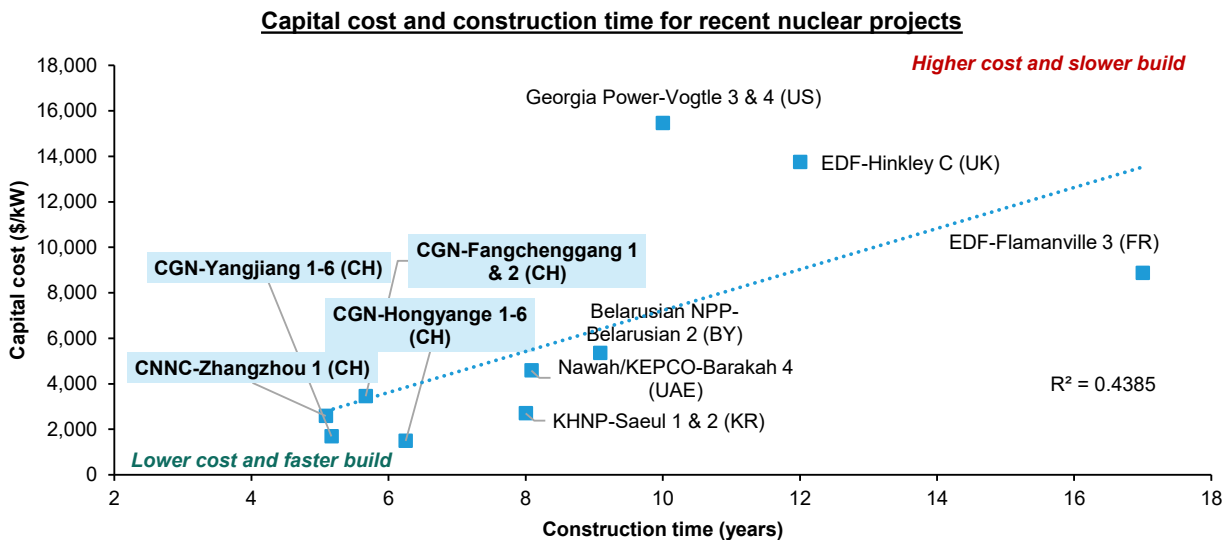
EXHIBIT 19: Cost comparison with the US. Costs are significantly lower in China which is key to hit net zero target in the future



Solar: Price of monocrystalline silicon module (c-Si); Wind: Price by signing date (excluding installations); Battery: Blended battery pack price; Electrolyzer: Electrolysis system capex cost (Alkaline).
Source: BNEF, Bernstein analysis and estimates

China's power advantage extends beyond low-cost renewables and battery storage. The country has also built one of the world's most competitive nuclear industries, with recent Chinese nuclear projects typically completed in 5-7 years at capital costs of roughly US\$2,000-2,500/kW, substantially below major western projects such as Vogtle 3&4 in the US, Hinkley Point C in the UK and Flamanville 3 in France. This combination of lower construction costs and faster deployment provides China with a scalable source of 24/7 low-carbon baseload power that complements intermittent wind and solar generation. For AI data centers, which require highly reliable and continuous electricity supply, competitively built nuclear power strengthens grid stability while helping support long-term electricity affordability, further reinforcing China's structural advantage in powering large-scale AI compute.

EXHIBIT 20: Western nuclear projects have exceeded 6,000/kW. Chinese nuclear plants are built for approximately 2,000-2,500/kW



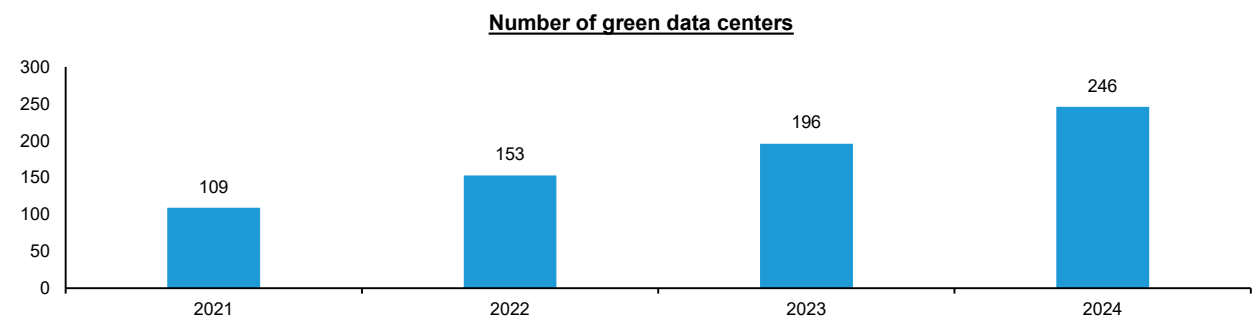
Source: Bloomberg, Company data, Bernstein analysis

POLICY PUSH FOR GREEN DATA CENTERS

For AI workloads, green data centers are particularly important because electricity has become one of the largest operating costs for hyperscale computing. As AI training and inference demand continue to grow, China is seeking to build a data center

ecosystem that combines low-cost power, high efficiency and a high share of clean energy. China has been steadily increasing the number of certified green data centers, rising from 109 in 2021 to 246 in 2024. The rapid growth reflects Beijing's view that AI and digital infrastructure should be developed alongside energy efficiency and decarbonization goals rather than through unconstrained power consumption.

EXHIBIT 21: **China has been steadily increasing the number of certified green data centers, rising from 109 in 2021 to 246 in 2024**



Source: China National Data Administration, Oxford Institute for Energy Studies, Bernstein analysis

China has introduced a series of policies designed to integrate data-center development with energy infrastructure planning. The 2022 "East Data, West Computing" initiative directed new large-scale data centers toward designated national computing hubs while encouraging latency-insensitive workloads such as AI training and batch processing to migrate to western provinces. The objective is to colocate compute demand with abundant and lower-cost energy resources. Subsequent policies have increased the focus on efficiency and renewable energy. The 2024 Green Low-Carbon Data Center Action Plan requires new national hub data centers to achieve more than 80% green power consumption, while tightening efficiency targets to PUE below 1.25 for new large data centers and below 1.20 for national hub projects. More recently, China has promoted green power direct-supply projects and "Computing-Power Grid Coordination" (算电协同), which seeks to integrate computing capacity, renewable generation, storage and grid infrastructure into a unified planning framework. Together, these policies show that China is not simply expanding data center capacity but increasingly coordinating compute growth with the power system.

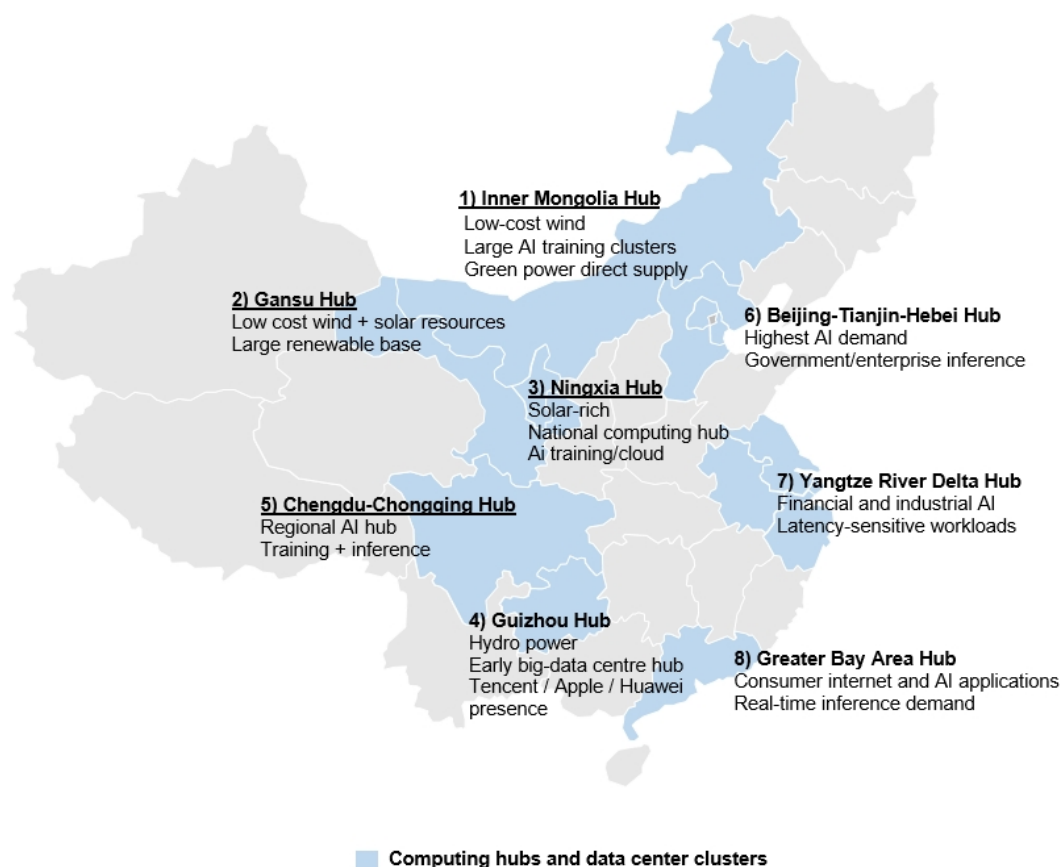
EXHIBIT 22: **China's data-center policies align compute growth with power availability, efficiency and cost**

Policy	Date of release	Requirement
Eastern Data Western Computing	Feb-22	- Layout rigidity: new large DC prioritize hub clusters; - Non-real-time load migrate west; - By end-2025 hub nodes $\geq 60\%$ of national new compute
Green Low-Carbon DC Action Plan	Jul-24	- New national hub DCs green power consumption >80% - Average DC PUE <1.5 - New large DCs PUE <1.25 - National hub DCs PUE <1.20
Green Power Direct Connection	2025	- Allows renewable generation projects to supply power directly to nearby industrial loads - Supports source-grid-load-storage projects - Facilitates dedicated green-power supply for energy-intensive users including datacenters
Computing-Power Grid Coordination	2025/26	- New national computing hubs must reach $\geq 80\%$ green power penetration; mandated into official govt agenda - General DCs: 10-15% storage ratio, 2hr duration. AI-focused DCs: 15-20% storage ratio, 2-4hr duration (based on industry commentary)

Source: Government data, Bernstein analysis

A key pillar of China's AI and data-center strategy is the "East Data, West Computing" program, which established eight national computing hubs across the country. The western hubs, including Inner Mongolia, Ningxia, Gansu and Guizhou, are located in regions with abundant renewable resources, lower electricity costs and greater land availability. These locations are primarily intended for AI training, large-scale data processing and other computing workloads that are less sensitive to latency. Meanwhile, eastern hubs such as Beijing-Tianjin-Hebei, the Yangtze River Delta and the Greater Bay Area remain close to major population centers and enterprise customers, making them more suitable for real-time AI inference and cloud applications. By relocating power-intensive computing workloads to resource-rich western regions and leveraging China's UHV transmission network, the country can reduce computing costs, improve renewable-energy utilization and alleviate power constraints in densely populated coastal provinces.

EXHIBIT 23: By relocating power-intensive computing workloads to resource-rich western regions and leveraging China's UHV transmission network, the country can reduce computing costs, improve renewable-energy utilization and alleviate power constraints in densely populated coastal provinces



Source: Wikicommon, Bernstein analysis

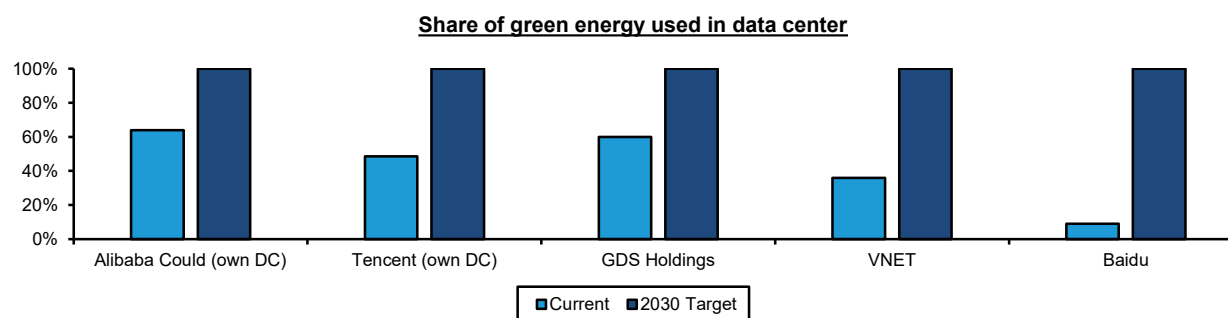
China's leading cloud and data center operators have adopted increasingly ambitious renewable energy targets. Alibaba Cloud and Tencent have both committed to achieving 100% clean or green energy by 2030, while other major operators including ByteDance, GDS, VNET, AtHub and Chindata have announced similar renewable-energy goals. Several operators have already made substantial progress, with Alibaba reporting clean power penetration above 60% for self-built data centers and Tencent reporting renewable-energy usage above 80% at owned facilities. Independent data-center operators are also accelerating renewable-energy procurement. GDS reported renewable-energy usage of around 60% in 2025, while VNET's renewable-energy consumption doubled from the prior year and exceeded 80% at some campuses.

EXHIBIT 24: China's leading cloud and data center operators have adopted increasingly ambitious renewable energy targets

Operator	Target level	Target year	Progress disclosed
Alibaba Cloud / Alibaba Group	100% Clean energy	2030	2025: self-built DC clean elec 64.0% (PUE 1.190); leased 22.4% clean; group clean elec share 52.2%
Tencent	100% Green energy	2030	2025: group green elec 48.5% (from 22% in 2024); owned DC RE 82.9%; leased DC RE 21.4%; purchased 4.13 TWh green
ByteDance	100% RE	2030	Limited granular public RE% in open sources reviewed; commitment tracked in Clean Cloud 2024
GDS Holdings	100% RE	2030	2025: RE usage 60%; direct green power purchases +60% YoY (75% of RE sources); avg PUE 1.23
VNET (21Vianet)	100% RE	2030	2025: RE 36% (1.25 TWh); was 18% in 2024; PUE 1.24; Ulanqab campus >80% RE
Shanghai AtHub	100% RE	2030	Commitment listed Clean Cloud 2024; detailed progress % not independently verified here
Chindata Group	100% RE	2030	Early RE leader in GP rankings (historical); post-privatization disclosure may vary
Baidu	Carbon neutral	2030	RE ratio lagged peers in GP 2024 (~low single-digit % cited for 2023); commitment on carbon neutrality

Source: Company data, Bernstein analysis

These commitments are being driven by both policy requirements and economic incentives. As green power becomes increasingly available through direct renewable procurement, green-power trading and dedicated renewable projects, operators can simultaneously lower carbon emissions, improve regulatory compliance and potentially reduce long-term electricity costs. As AI electricity demand grows, renewable-energy procurement is expected to become an increasingly important competitive differentiator across China's cloud and data-center industry.

EXHIBIT 25: Alibaba Cloud and Tencent have both committed to achieving 100% clean or green energy by 2030, while other major operators including ByteDance, GDS, VNET, AtHub and Chindata have announced similar renewable-energy goals


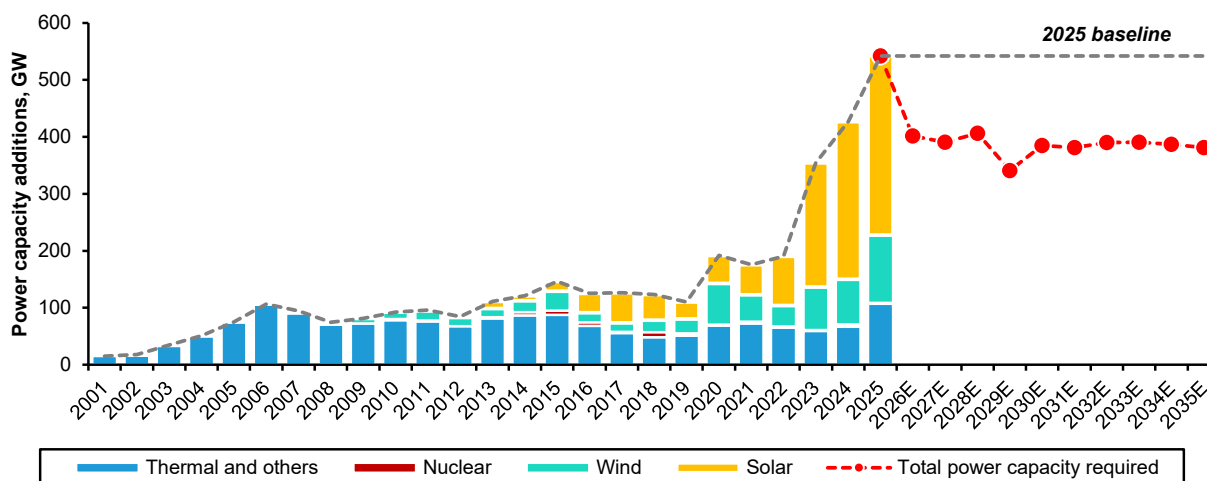
Source: Company data, Bernstein analysis

GRID AND STORAGE EXPANSION ARE ESSENTIAL TO SUPPORT AI GROWTH

To meet China's rapidly rising electricity demand, total installed power generation capacity will need to grow by approximately 385GW per year, representing a 7.1% CAGR over the coming decade. This implies an expansion from 3,891GW in 2025 to around 7,745GW by 2035. It also implies an increase in annual power capacity significantly higher than the 50GWh needed in the US. China's demonstrated ability to scale capacity additions supports the feasibility of this trajectory. In 2025 alone, the country added more than 500GW of new generation capacity, driven largely by accelerated deployment of solar and

wind projects alongside continued investments in grid infrastructure. This remarkable build-out indicates that China already possesses the construction capacity, supply chain depth, and policy alignment required to consistently deliver 350-400 GW of new capacity per year going forward. Such sustained growth will be essential not only to satisfy rising power consumption but also to enable China's broader energy transition goals, including electrification of industry and transport, integration of renewable energy, and eventual decarbonization of the power sector.

EXHIBIT 26: China added more than 500GW of power capacity in 2025, indicating its capability to cover the following power demand of 350-400GW annually

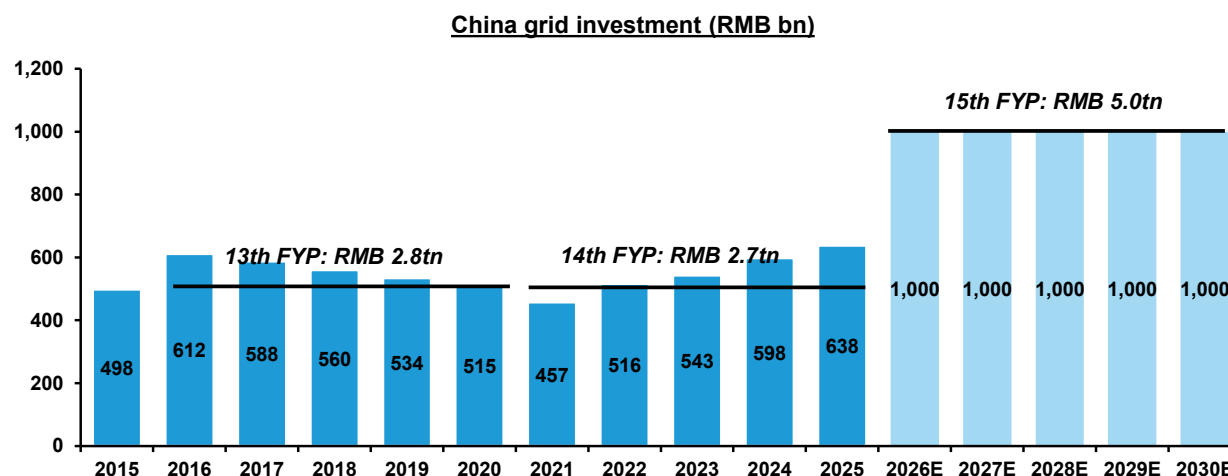


Forecasts are based on the average capacity factor of new generation added between 2024 and 2025 (13.6%)

Source: China NBS, Bernstein analysis and estimates

China's ability to convert low-cost western generation into usable eastern computing capacity depends heavily on its ultra-high-voltage (UHV) transmission network. By mid-2026, China had 43 UHV transmission lines in operation. The UHV system effectively allows China to operate as a single national power market, moving energy-intensive AI workloads to areas where renewable power is cheapest rather than forcing datacentres to compete for limited electricity near demand centres.

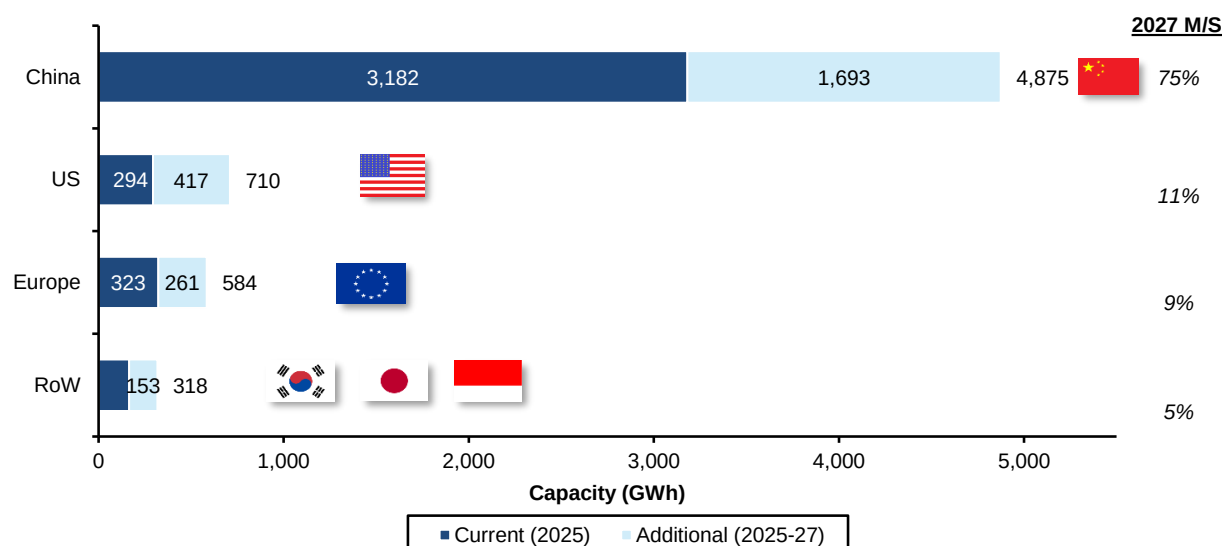
China plans to invest more than RMB5tn in grid infrastructure during 2026-30. The country also aims to increase West-to-East transmission capacity to more than 420GW, add 15 new UHV DC transmission lines, and build a grid capable of integrating over 2.8TW of wind and solar capacity by 2030. Without substantial investment in transmission infrastructure, renewable curtailment would increase and renewable deployment would ultimately slow.

EXHIBIT 27: China plans to invest more than RMB5tn in grid infrastructure during 2026-30

Source: IEA, Bernstein estimates and analysis

Battery storage will be another critical factor in determining how quickly renewable energy can be scaled to meet rising power demand. Energy storage is needed for storing power when the sun doesn't shine or when the wind doesn't blow. With the integration of large-scale solar and wind power, the grid will require more storage to deal with under supply and curtailment of power. Battery storage will help improve the stability of power supply and help with peak regulation and frequency regulation.

As with the development in any other renewables related industries, energy storage is another industry that China dominates. China enjoys an 80% market share of the Li-ion battery market. China's battery manufacturing capacity reached >3,000GWh at the end of 2025, more than 4x the RoW put together. Given the strategic importance of batteries in electrification of energy markets and transport, this gives China a competitive advantage in an industry which will reach US\$1trillion in sales by 2050.

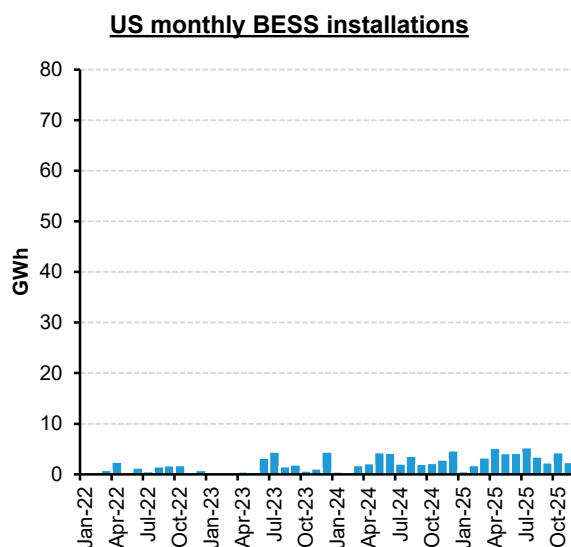
EXHIBIT 28: Global battery capacity (2025 and 2027) by region. China will continue to dominate but US and Europe will be fastest growing regions

Source: Company data, Bernstein estimates and analysis

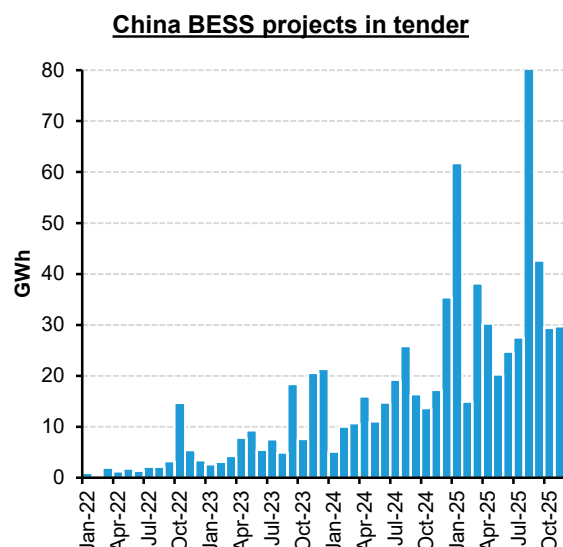
Last year was a breakout year for battery energy storage (BESS) market in both China and the U.S., reflecting surging demand driven by rapid renewable-energy deployment. In 2025, global BESS battery demand significantly exceeded expectations, growing by an estimated 85% y-o-y to reach 556GWh, which represents 30% of the total battery demand. ESS demand is

accelerating faster than anticipated, driven by significant reductions in solar and storage costs. Rising power consumption, such as fueled by data centers, electric vehicles, and air conditioning, further amplifies the need for energy storage.

We expect China to install around 300GWh of BESS capacity this year (+95% y-o-y), compared with 70-80GWh in the U.S. (+40% y-o-y). These installation levels are broadly aligned with the scale of renewable power generation in each market (approximately 2,300TWh in China versus 760TWh in the U.S.), but they are also heavily influenced by local supply availability. While China's domestic battery manufacturing capacity appears sufficient to support rapid BESS growth, the U.S. market remains supply-constrained in the near term. U.S. ESS manufacturing capacity is expected to rise from 36GWh at end-2025 to 102GWh by end-2026, but this still lags projected demand.

EXHIBIT 29: **The U.S. ESS battery demand**

Source: EIA, Bernstein analysis

EXHIBIT 30: **China ESS battery demand**

Source: S&E Consulting, Bernstein analysis

STOCK IMPLICATIONS

For investors looking to position around China's AI-driven power buildout, the opportunity set spans three key areas: grid infrastructure, power generation, and energy storage. Grid equipment and UHV transmission suppliers should be among the most direct beneficiaries of State Grid and China Southern Power Grid's planned investment of more than RMB1tn annually over the next five years, as China expands transmission capacity to connect renewable-rich western regions with major computing hubs. Renewable power and energy storage providers should also benefit from policy requirements that new national computing hubs source at least 80% of their electricity from green power and 15-20% of energy storage. Energy storage is becoming an increasingly critical component of the AI ecosystem as policymakers promote greater integration of computing, power and the grid. AI data centers are also expected to deploy higher levels of storage capacity to improve power reliability and support renewable integration.

EXHIBIT 31: **Beneficiaries of China's electrification mega trend**

Power Equipment & New Energy	Electrical Equipment	Power Electronics & Automation	Inovance	Nari	Megmeet	Hongfa	CSPG Tech
		Transmission & Distribution Equipment	TBEA	Sieyuan	Ningbo Orient	China XD	State Grid Yingda
		Distribution Equipment	CHNT	Sanxing Medical	Nader	Sineng Electric	
		Motors	MOONS	Fortior Tech	Jiangsu Leili	Roshow	Keli Motor
	Power Equipment	Solar Energy	Sungrow	Longi	Tongwei	Jinko Solar	Daqo
		Integrated Energy Equipment	Shanghai Electric	Dongfang Electric	CSIC	Wuxi Huaguang	ZHEFU
		Wind Power	Goldwind	Sany	Ming Yang	Shanghai Electric Wind	Xiangtan
		Energy Storage	Kehua Data	Kstar	Hyperstrong	Narada	Camel
		Nuclear Power	Sufa	CGN Nuclear Tech			
	New Energy Power Systems	Lithium Batteries	CATL	EVE Energy	Gotion	Putailai	Tianheng
		Automotive Motor Control	Wolong	Broad Ocean Motor	JSEM	VMAX	
		Battery Integrated Services	GEM	Tgood	Sinexcel		
		Fuel Cells	AT&M	MOON-Tech	Zhejiang Jiahua	Fujian Snowman	

Electric Power & Utilities	Generation & Grid	Hydropower	CYPC	Huaneng Hydropower	Sichuan Chuantou	Guangxi Guiguan	GEPIC
		Other Generation	GNNC	CGN Power	CTGR	Longyuan	Fujian Funeng
		Thermal Power	SDIC Power	Huaneng	GD Power Development	Zhejiang Zheneng	Huadian Power
		Grid	CSPGES	Fuling	Chongqing Three Gorges	Leshan Electric Power	

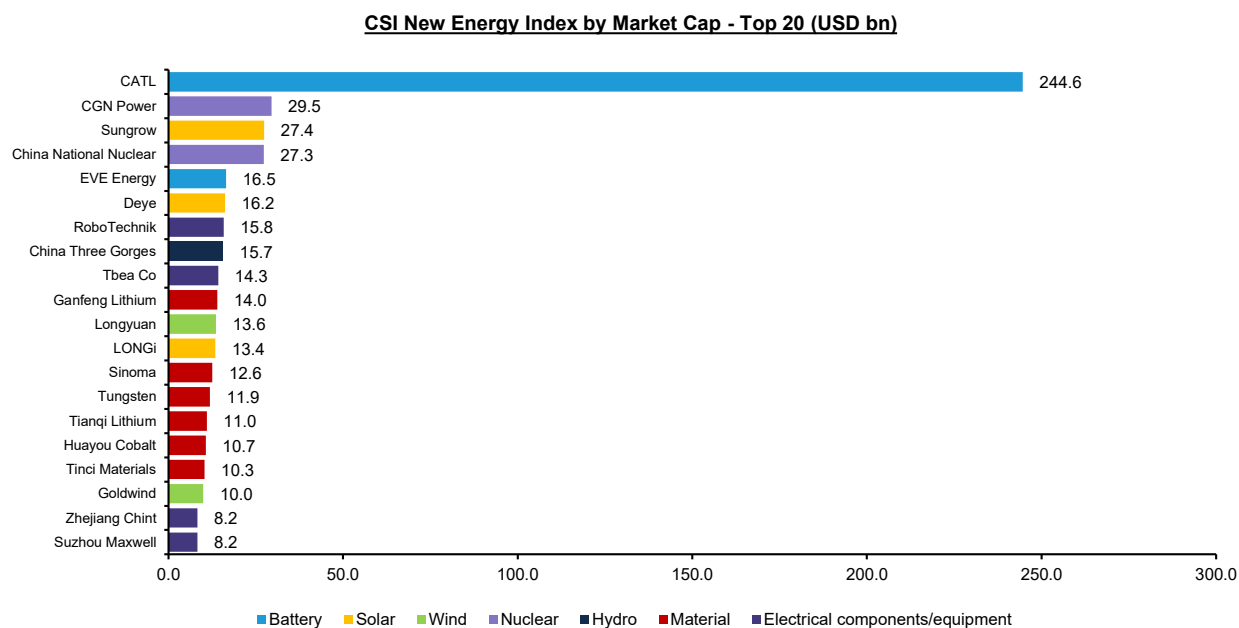
Bubble size represents total market cap as of September 14, 2026.

CATL and Sungrow is covered by Bernstein. All other stocks are not covered by Bernstein

Source: Bloomberg, Bernstein analysis

Among our coverage, **CATL** is a key beneficiary given its leadership in battery technology and grid-scale storage systems, while **Sungrow** is well positioned through its exposure to power conversion equipment, renewable integration, energy storage systems and grid infrastructure. Together, both companies offer direct exposure to the electrification of AI and power infrastructure.

EXHIBIT 32: **CSI New Energy Index by market capitalization, which will likely benefit from the growth in power demand**



As of September 14, 2026

Source: Bloomberg, Bernstein analysis

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